Recurrences I

COMS10018 - Algorithms

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Divide-and-conquer Algorithms

Algorithmic Design Principle: Divide-and-conquer

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Examples

Quicksort, Mergesort, maximum subarray algorithm, binary search, FAST-PEAK-FINDING, ...

Example: Mergesort

Recall: Mergesort

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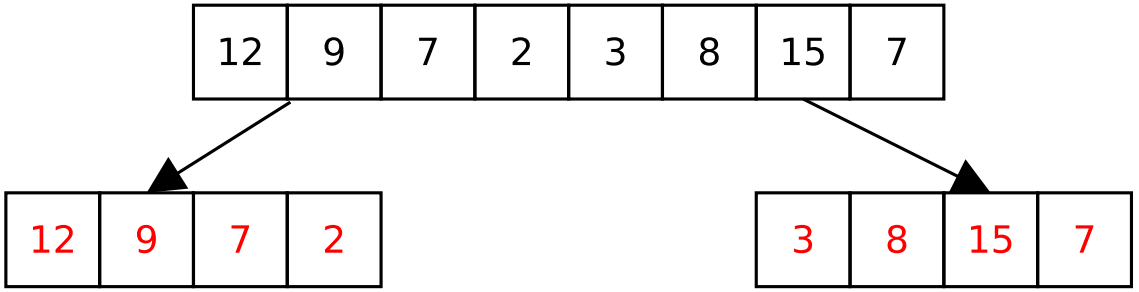
Recall: Mergesort

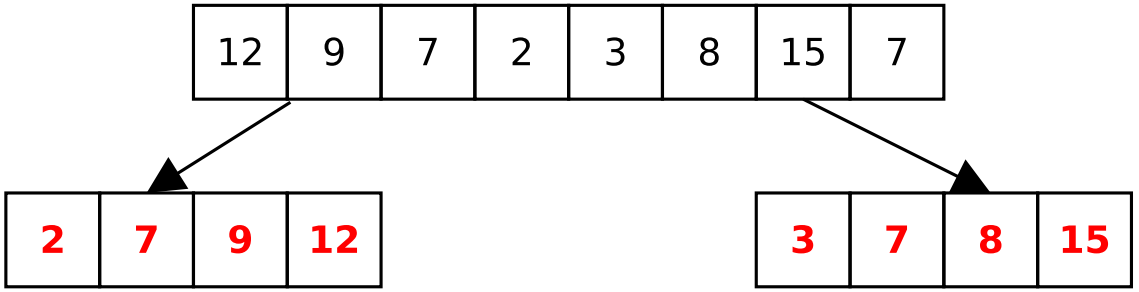
1. **Divide**
   Split input array A of length n into subarrays $A_1 = A[0, \lfloor n/2 \rfloor]$ and $A_2 = A[\lfloor n/2 \rfloor + 1, n - 1]$



Example: Mergesort

Recall: Mergesort

1. **Divide** $A \rightarrow A_1$ and A_2
2. **Conquer**
   Sort A_1 and A_2 recursively using the same algorithm



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1. **Divide** $A \rightarrow A_1$ and A_2
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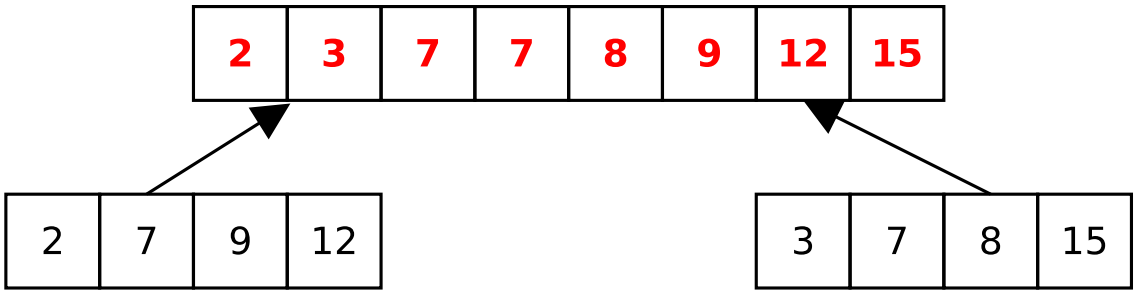
   Combine sorted subarrays A_1 and A_2 and obtain sorted array A



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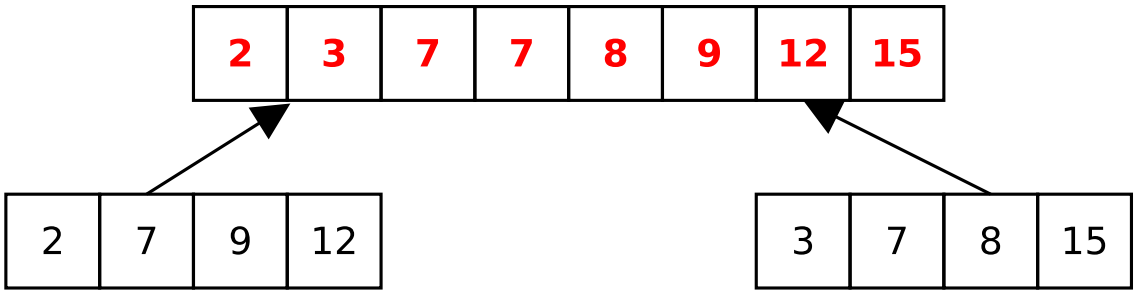

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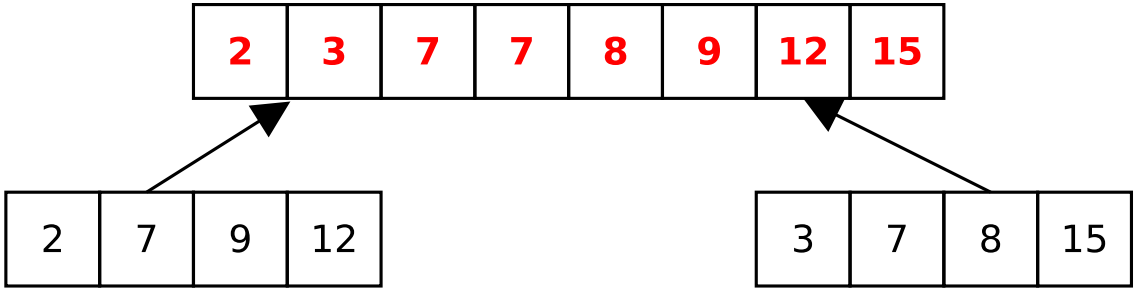

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$$T(1) = O(1)$$

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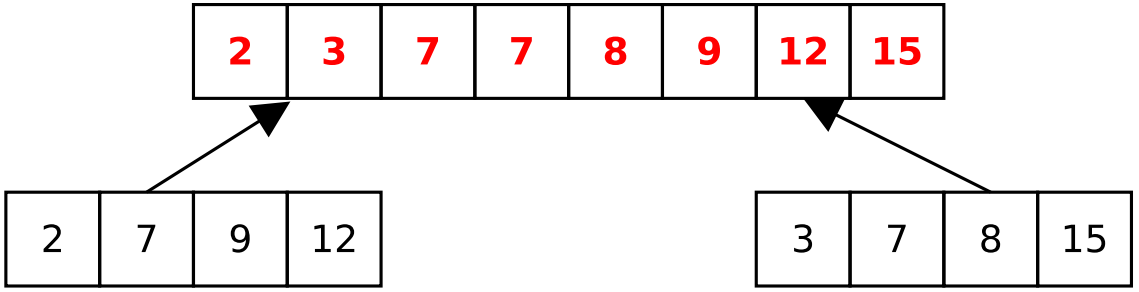

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  very powerful, cannot always be applied

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The Substitution Method (2)

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The base case is a problem...

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Recall: $T(1) = c_1$ and $T(n) = 2T(n/2) + c_2 n$

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- We proved $T(n) \leq Cn \log n$, for every $n \geq 2$, when choosing $C \geq c_1 + c_2$

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- We proved $T(n) \leq Cn \log n$, for every $n \geq 2$, when choosing $C \geq c_1 + c_2$
- **Observe:** This implies $T(n) \in O(n \log n)$ (important)

The Substitution Method (3)

Recall: $T(1) = c_1$ and $T(n) = 2T(n/2) + c_2 n$
Our guess: $T(n) \leq Cn \log n$ (induction step holds for $C \geq c_2$)

Solution: Choose a different base case! $n = 2$

$$\begin{aligned} T(2) &\leq 2T(1) + 2c_2 = 2c_1 + 2c_2 = 2(c_2 + c_1) \\ C2\log 2 &= 2C \end{aligned}$$

Hence, for every $C \geq c_2 + c_1$, our guess holds for $n = 2$:

$$T(2) \leq C2\log 2 \ .$$

Result

- We proved $T(n) \leq Cn \log n$, for every $n \geq 2$, when choosing $C \geq c_1 + c_2$
- **Observe:** This implies $T(n) \in O(n \log n)$ (important)

Asymptotic notation allows us to chose arbitrary base case

A Strange Problem

A Strange Problem

Example: Give an upper bound on the recurrence

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$$\begin{aligned} T(1) &= 1 \\ T(n) &= T(\lceil n/2 \rceil) + T(\lfloor n/2 \rfloor) + 1 \end{aligned}$$

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Step 1: Guess correct solution

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$$T(n)$$

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Step 2: Verify the solution

$$T(n) \leq C\lceil n/2 \rceil + C\lfloor n/2 \rfloor + 1$$

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- We need a different guess

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- Let's try: $f_1(n) := Cn + 1$ and $f_2(n) := Cn - 1$

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$f_1 : T(n)$

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$$f_1 : T(n) \leq C\lceil n/2 \rceil + 1 + C\lfloor n/2 \rfloor + 1 + 1$$

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$$f_1 : T(n) \leq C\lceil n/2 \rceil + 1 + C\lfloor n/2 \rfloor + 1 + 1 = Cn + 3$$

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- Let's try: $f_1(n) := Cn + 1$ and $f_2(n) := Cn - 1$

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$$\begin{aligned} f_1 : T(n) &\leq C\lceil n/2 \rceil + 1 + C\lfloor n/2 \rfloor + 1 + 1 = Cn + 3 \not\leq f_1(n) \text{ ✗} \\ f_2 : T(n) & \end{aligned}$$

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(holds for every positive C)

A Strange Problem (2)

Verify Base Case for f_2

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- We have: $T(1) = 1$ and $f_2(1) = C - 1 \geq T(1)$ for $C \geq 2$

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Comments

A Strange Problem (2)

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- Guessing right can be difficult and requires experience

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Comments

- Guessing right can be difficult and requires experience
- However, recursion tree method can provide a good guess!